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# FusionMAE, a self-supervised pretrained model to optimize and simplify diagnostic and control of fusion plasma

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In magnetically confined fusion device, the complex, multiscale, and nonlinear dynamics of plasmas necessitate the integration of extensive diagnostic systems to effectively monitor and control plasma behaviour. The complexity and uncertainty arising from these extensive systems and their tangled interrelations have long posed a significant obstacle to fusion energy development. In this work, a self-supervised model, fusion masked auto-encoder (FusionMAE) is pre-trained to compress the information from 88 diagnostic signals into a concise embedding, to provide a unified interface between diagnostic systems and control actuators. Two mechanisms are proposed to ensure a meaningful embedding: compression-reconstruction and missing-signal reconstruction. Upon completion of pre-training, the model acquires the capability for 'virtual backup diagnosis', enabling the inference of missing diagnostic data with 97.2% accuracy. Furthermore, the model demonstrates multiple downstream applications: automatic data analysis, universal control-diagnosis interface, and enhancement of control performance on multiple tasks. This work pioneers the integration of large-scale artificial intelligence (AI) models into the field of fusion energy. It demonstrates that pre-trained embeddings can simplify the system interface, reduce diagnostic redundancy and optimize operation performance, paving the way for high-performance future fusion reactors.

Fusion energy has long been regarded as a clean and sustainable power source, with the potential to meet the rapidly growing global energy demand while addressing carbon neutrality. Among the various concepts, tokamak stands out as one of the most promising designs for realizing the first commercial fusion reactor, having achieved remarkable milestones in recent years. For instance, the Joint European Torus set a world record by producing 59 megajoules of fusion energy over a 5-second pulse[1]. The Korea Superconducting Tokamak Advanced Research sustained plasma with ion temperatures exceeding 100 million kelvin for 30 seconds[2]. And the EAST and WEST tokamak realized a steady state plasma lasting up to 1000 and 1320 s, respectively[3]. Meanwhile, the International Thermonuclear Experimental Reactor (ITER), one of the world's largest science projects, involving the collaboration of more than 30 nations, is currently under assembly to demonstrate the scientific feasibility of tokamak reactors[4].

Despite the remarkable achievements, the path to a Fusion Power Plant (FPP) is still hindered by several physical and engineering challenges[5]. Controlling and optimizing fusion plasma, a non-linear magnetohydrodynamic system, requires the integration of highly intricate, multidisciplinary subsystems[6]. A wide range of diagnostic systems have been developed to measure plasma properties, including current, temperature, density, rotation, and radiation[7]. Additionally, numerous simulation modules have been created to model plasma behaviour across different time scales and spatial regions[8]. Furthermore, a significant number of control actuators have been designed to regulate plasma properties[9]. For instance, ITER incorporates roughly 50 distinct plasma diagnostic systems and 70 sets of control actuators[10]. Moreover, the highly complex Integrated Modelling & Analysis Suite (IMAS) system has been established to support experimental data analysis and simulation program execution[11]. In parallel, various control loops are formed through combining different modules, working in coordination to achieve the overall operation of the fusion reactor[12]. Therefore, while the scientific feasibility of fusion energy has been validated, the engineering complexity of fusion power plant remains prohibitively high, as reflected in the extremely long timeline of ITER[13].

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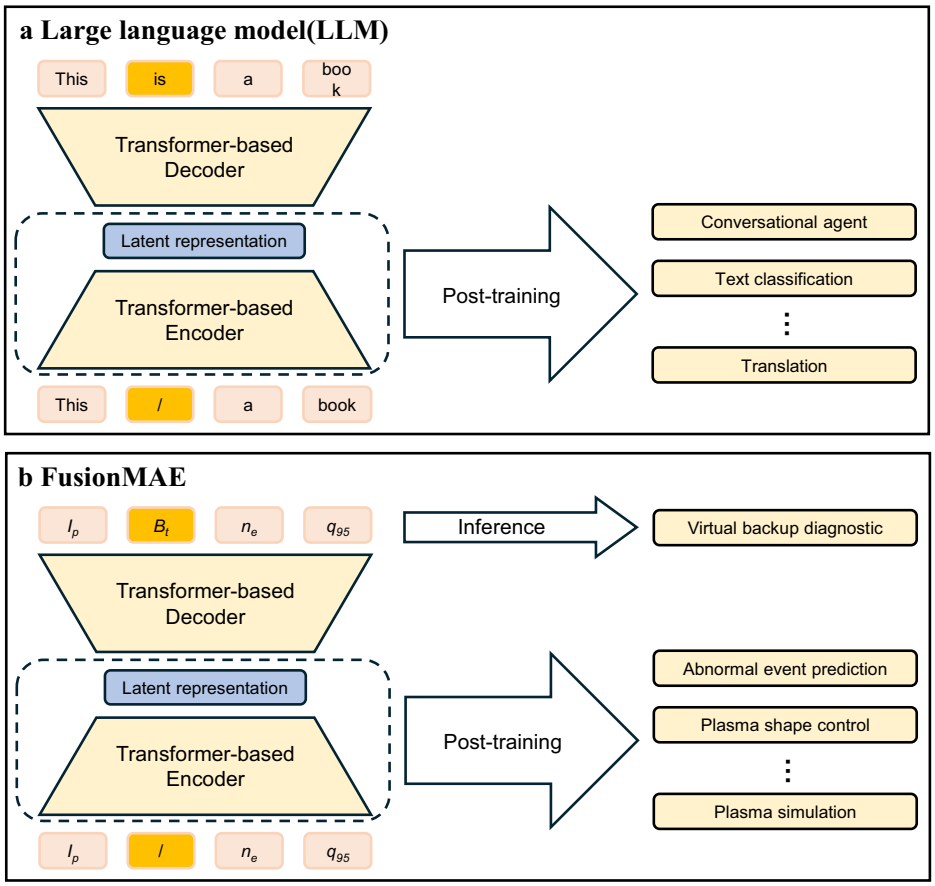



**Fig. 1 | Comparison between the pre-training and downstream application frameworks of large language models and FusionMAE. a** The Transformer-based encoder-decoder model is pre-trained by predicting masked words(self-supervised task) or predicting next word(self-regression task). The encoder then produces a latent representation rich in contextual information, which can be post-trained for various downstream tasks, including conversational AI, text classification, and machine translation. **b** The Transformer-based encoder-decoder model is pre-trained by reconstruct missing diagnostic data. The encoder then produces a latent representation for plasma state, which can be post-trained for abnormal event prediction, plasma shape control and plasma simulation. The whole encoder-decoder model can also be used as virtual backup diagnostic in fusion devices.

In recent years, artificial intelligence (AI) technologies have been widely applied in computer vision, natural language processing(NLP), and diverse scientific fields such as biology, materials science, and meteorology[14–18]. AI has proven its capability in providing end-to-end, streamlined solutions for complex problems. In the fusion domain, AI has also demonstrated numerous successful applications, including predicting major plasma disruptions[19,20], automating the realization of complex plasma magnetic configurations[21], and achieving kinetic control to avoid plasma instabilities[22]. While these specialized models have made valuable contributions to specific challenges in fusion research, particularly in localized decision-making and control of certain plasma characteristics, their application scope remains limited. These models primarily focus on understanding specific aspects of plasma physics related to the downstream tasks, which may not provide a comprehensive grasp of the underlying principles. In contrast, a paradigm represented by large language models (LLMs) proposes that, by developing a pretrained model that comprehensively understand plasma physics and use fine-tune, post-train or prompt engineering techniques to turn the pretrained model into task-specific scenarios, the performance on downstream tasks could be further promoted. Furthermore, within the base model-downstream task framework, solutions to various downstream tasks can be unified into a common algorithmic workflow. This approach would significantly simplify the control systems of fusion devices and allow for more efficient utilization of diagnostic information. This unified framework could substantially improve the overall efficiency of fusion device operation, enhancing both control precision and the optimization of key plasma characteristics.

In this work, we present a pretrained model named the Fusion Masked Auto-Encoder (FusionMAE), which compresses data from multiple diagnostic systems into a unified plasma state embedding, based on a dataset from the largest tokamak currently operated in China, HL-3[23,24]. This embedding allows a Transformer[25] decoder to restore all the raw diagnostic data and reconstruct possible missing data from it, showing a successfully integration of various diagnostic signals. Analogous to word embeddings in NLP[26], the plasma state embedding naturally clusters discharges according to similarities in operational scenarios. FusionMAE demonstrates extensive utility as a foundational model for HL-3, including leveraging its original capacity to impute missing diagnostic data for the development of virtual backup diagnostics, as well as adapting its masking mechanism to enable automated secondary data analysis. Consistent with our expectations for a pretrained model, it can support multiple downstream tasks, disruption prediction (DPR), equilibrium reconstruction (EFIT-NN) and plasma evolution prediction (PPR)[27–29]. The performance of these task-specific models can all benefit from FusionMAE and keep steady even with the failure of several input diagnostics. Figure 1 compares the pre-training and downstream application frameworks of FusionMAE and LLMs. It demonstrates that FusionMAE is positioned as a device-specific pretrained model designed for fusion experimental data, holding significant potential to evolve into a general-purpose agent tailored to the fusion research.

## Results

### FusionMAE: a Transformer-based model pretrained by self-supervised learning task

FusionMAE is developed to compress 88 signals from 12 systems of HL-3 into a unified embedding. The input consists of plasma current ($I_p$), plasma shape and position ($R$, $Z$, $a$, $\kappa$, $\delta_u$, $\delta_l$), toroidal field ($B_t$), poloidal field ($B_p$), loop voltage ($V_{loop}$), electron density ($n_e$), electron temperature ($T_e$), stored energy ($W_e$), radiation power ($P_{rad}$), soft-X radiation ($SX$), impurity radiation($A_{imp}$), D-α spectrum ($D_\alpha$), amplitude of magnetohydrodynamic instability modes ($A_{MHD,\ 1}$, $A_{MHD,\ 2}$), normalized beta($\beta_N$), safety factor($q_{95}$), internal inductance($l_i$), poloidal field coil current ($I_{PF}$) and auxiliary heating power ($P_{NBI}$, $P_{EC}$, $P_{LH}$). The diagram of these diagnostic and control systems are given in Fig.2a, The auxiliary heating systems are omitted due to its complexity, while the details can be found in ref. 30. Additionally, some secondary parameters, like $A_{MHD}$, $W_e$, $\beta_N$, $q_{95}$ and $l_i$ are computed from a set of primary data using dedicated programs, as described

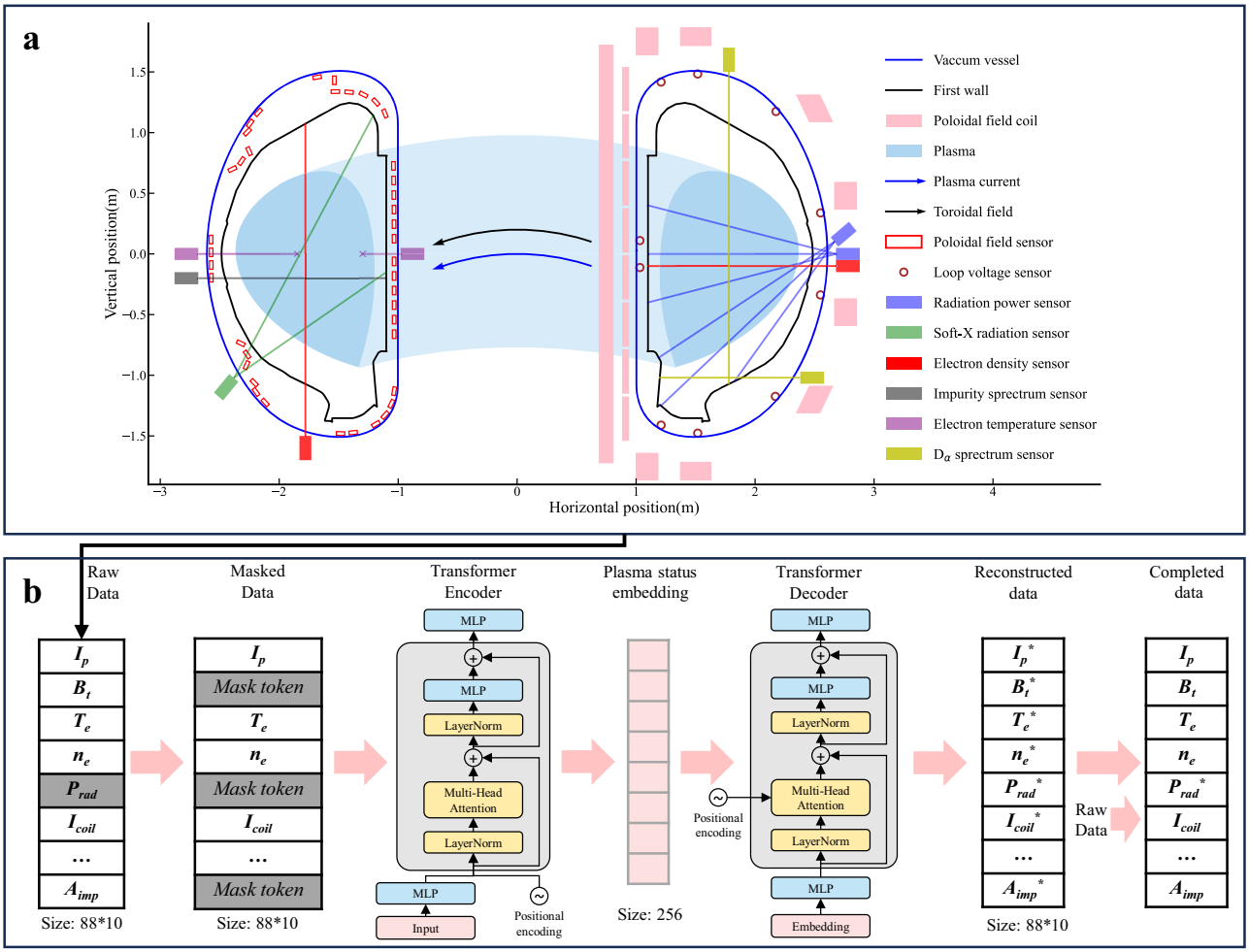



**Fig. 2 | Overall architecture of FusionMAE. a** Diagram of the diagnostic systems employed in this study. **b** Workflow and neural network structure of FusionMAE. The input signals are organized into a 2-D matrix, with missing signals and intentionally masked signals replaced by masked tokens. The encoder compresses the data into a latent plasma state embedding, and the decoder reconstructs the original signals, including the masked ones.

in refs. 31,32. Detailed descriptions of these signals are provided in the Methods section. These systems span a broad spectrum of physical parameters and can be considered as a representative subset of the systems required by future fusion reactors like ITER[10].

Figure 2b illustrates the workflow and neural network architecture of FusionMAE. It takes a 10 ms time window of the 88 signals sampled at 1 kHz as input, forming an $88 \times 10$ array. Firstly, each channel of the array is projected into a 128-dimensional space using a multilayer perceptron (MLP), and a positional encoding vector[33] is added to encode the channel index. Secondly, the resulting $88 \times 128$ array is fed to an encoder composed of several Transformer blocks, each consisting of multi-head self-attention layers[34], MLPs, layer-normalizations[35], and residual connections[36]. The encoder takes the main responsibility to learn underlying patterns of plasma state. The output of the Transformer encoder is further embedded by an MLP layer into a vector of length 256, which is the expected plasma state embedding. The decoder has a mirrored structure, where several Transformer blocks reconstruct the original data from the embedded vector. The only difference is that the positional embedding is not added and instead serves as the query vector in the final multi-head attention layer. The entire encoder-decoder structure is trained to minimize the mean squared error between the output and input signals.

### Unified plasma state embedding

FusionMAE extracts a meaningful embedding through two key mechanisms. Firstly, the embedded vector is designed to be smaller than the input array to force effective compression. To preserve information, the model must capture the intrinsic low-dimensional manifold of plasma dynamics. As shown in Fig. 3a, a black radar plot visualizes the Pearson correlation coefficients (PCC) between the reconstructed and original signals across all 88 channels. The averaged PCC for all channels is 98.7%, indicating that the embedding retains nearly all information contained in the raw diagnostic data, while providing a more compact and structured representation.

However, when the model is solely trained to compress and restore data, it tends to memorize signals on a channel-by-channel basis and struggles to capture the interconnections between channels. To address this limitation, a more challenging task has been devised. In each training step, a random 25% of the input channels are masked, forcing the model to reconstruct the channels based on its understanding of inter-channel relations. To validate FusionMAE's performance under this situation, two testing experiments are conducted. Firstly, we intentionally mask one of the channels, calculate the PCC between restored and original signals, and repeat the process across all the 88 channels. Secondly, we mask the signals from a same diagnostic or control subsystem together, calculate the PCC and repeat the process on 12 subsystems, which is a more stressful test. FusionMAE successfully achieved average PCCs of 97.2% and 95.3%, which is presented in Fig.3a by blue and red radar plots. FusionMAE shows a gradually declining performance in the three testing modes, without masking, with per-channel masking and with per-diagnostic masking, as the input information goes more incomplete. For $B_p$ signals, the PCC is very close to 100% with per-channel masking, showing that the corresponding

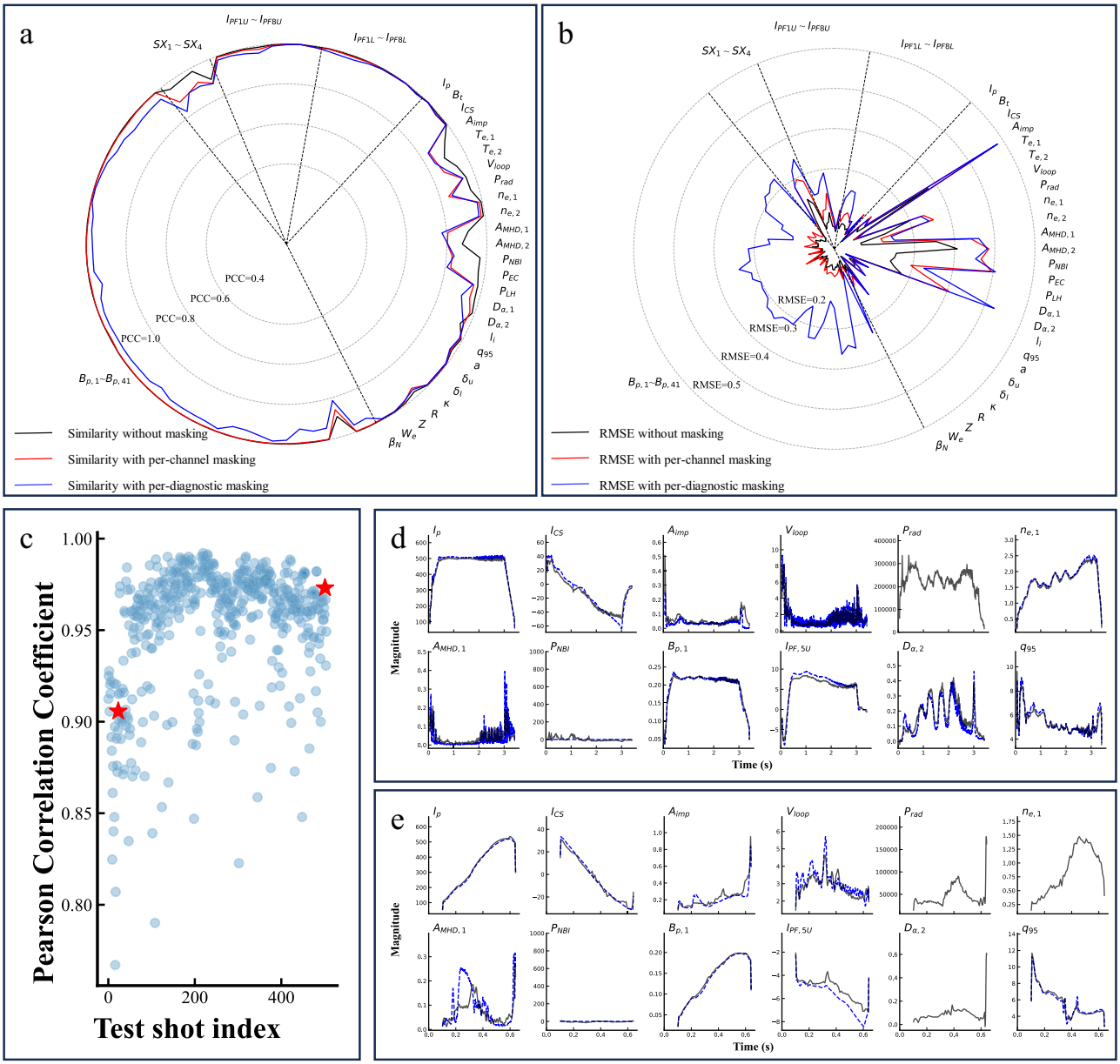



**Fig. 3 | Evaluation of FusionMAE's performance in restoring compressed raw signals and reconstructing missing signals. a** Radar plot illustrating the PCC between original signals and FusionMAE output across 88 channels. Gray radar plot is tested without masking. Red radar plot is tested by intentionally masking one of the 88 signals. Blue radar plot is a more stressed testing with all signals from a same diagnostic subsystem masked together. **b** Radar plot illustrating the normalized RMSE between original signals and FusionMAE output.**c** Scattered plot of averaged PCC with per-channel masking across 88 channels against the test shot index, ordered by the date and time of discharge. Red stars are representative shots in (**d**, **e**). **d** Comparison between reconstructed (black solid lines) and raw signals (blue dotted lines) for Shot 8909. Some raw signals are unavailable owing to operational anomalies in the diagnostic system.**e** Comparison between reconstructed (black solid lines) and raw signals (blue dotted lines) for Shot 1430. Some raw signals are unavailable owing to operational anomalies in the diagnostic system.

diagnostic is highly redundant and is not sensitive to the missing of a few signals. But there is an obviously larger performance degradation with per-diagnostic masking for $B_p$, compared to other diagnostics, which shows that the reconstruction of these signals will be as difficult as others when all the $B_p$ diagnostics fail. Figure 3b shows three radar plots for Rooted Mean Squared Error (RMSE) calculated on normalized data, which shows very similar results with Fig.3a. Further tests are also made on the data from the latest campaign of HL-3 and show similar results with subtle performance degradation, as shown in Supplementary Note 1.

Figure 3c shows the averaged PCC across the 88 channels for each test shot. For most of the shots, the averaged PCC is higher than 95%. Shots in early campaigns have a relatively lower PCC distribution, as the control system of HL-3 is still under development and the discharge is not very stable in this stage. Most shots with PCC lower than 90% are those that terminate early in the ramp-up phase and do not establish a steady plasma state, which may challenge the feasibility of FusionMAE. Figure 3d, e show examples of data reconstruction for Shot 8909 and Shot 1430, as representative cases with relatively higher and lower performance.

The plasma state embedding extracted by FusionMAE plays a role analogous to word embeddings in NLP. Represented by a 256-length vector, it encapsulates the intrinsic physical state of the plasma. Each diagnostic system provides a partial observation of this state, which is transformed into output either through the decoder in digital space, or by physical diagnostic hardware in real experiments. Therefore, this state should remain unchanged regardless of the number of diagnostic signals used, while more signals only refine the estimation, and fewer signals

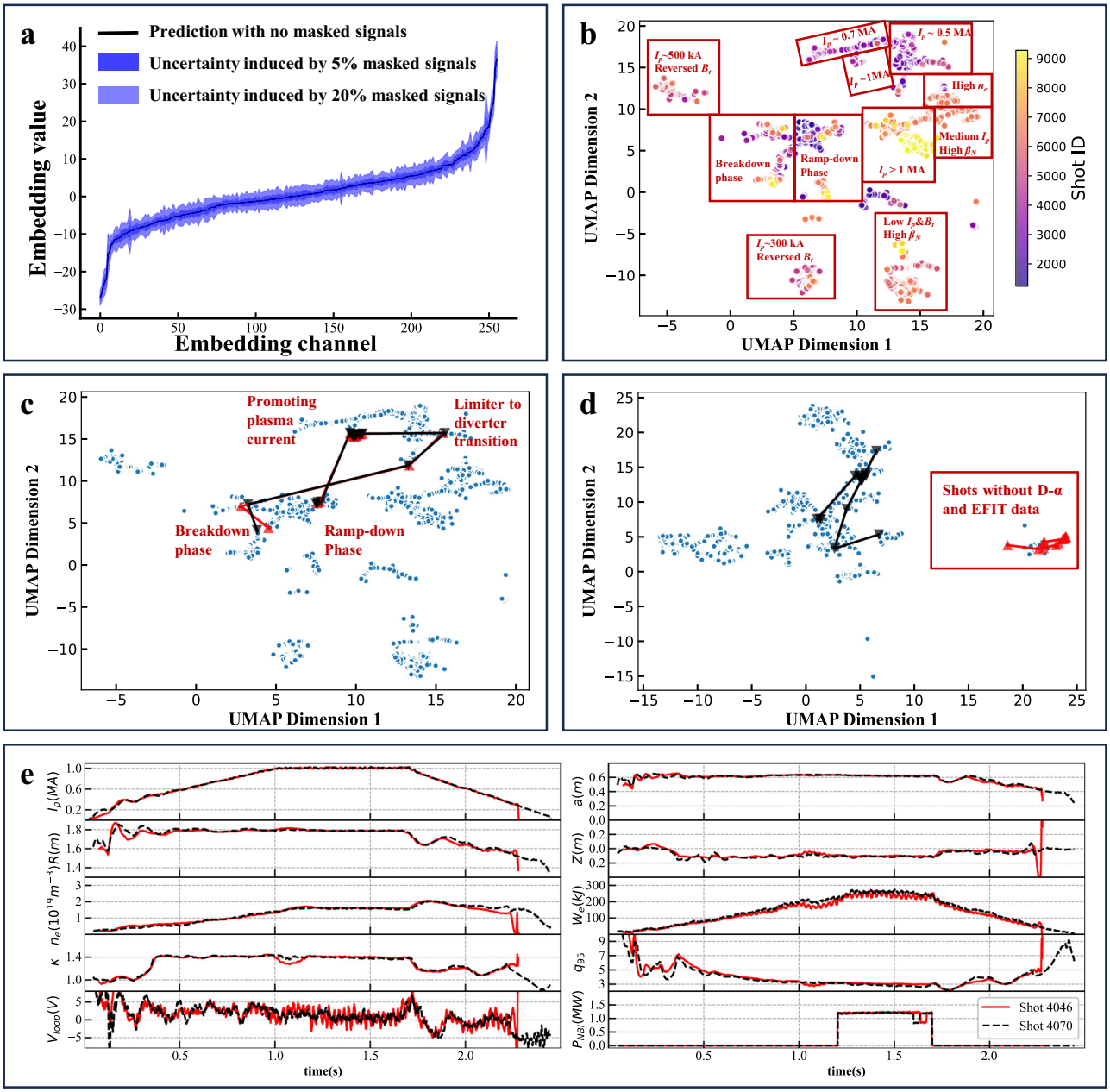



**Fig. 4 | Demonstration of a plasma state embedding and its similar characteristic as word embedding. a** A representative plasma state embedding generated by FusionMAE. Blue bands indicate the uncertainty induced by randomly masking 5% and 20% of the input signals. Embedding dimensions are sorted by value extracted from complete 88 signals for clarity. **b** UMAP projection of averaged embeddings from all 100 ms trajectories in the test set, revealing clustering aligned with toroidal field $B_t$, plasma current $I_p$, electron density $n_e$ and normalized beta $\beta_N$, which is marked by red boxes. **c** Trajectories of Shot 4046 and Shot 4070 in the UMAP plot generated from the embeddings. **d** Trajectories of the two shots in the UMAP plot generated from raw data. **e** Main plasma parameters during the two shots. Here the plasma shape parameters ($R$, $Z$, κ, …) for Shot 4046 are restored by FusionMAE.

result in a coarser approximation. This is demonstrated in Fig. 4a, when the diagnostic signals are randomly reduced by 5% and 20%, the extracted embeddings will only introduce limited oscillation around the black line. Moreover, the uncertainty induced by masking a given diagnostic can be used to evaluate the information contribution of that data, as demonstrated in Supplementary Note 2.

Analogous to word embeddings, the plasma embeddings of physically similar discharges are expected to cluster in latent space. This structure is confirmed via Uniform Manifold Approximation and Projection (UMAP) of embeddings from the test set as shown in Fig. 4b. Despite uniform treatment of all 88 input channels during training, FusionMAE autonomously identifies and organizes embeddings along key plasma parameters ($B_t$, $I_p$, $n_e$ and $\beta_N$), indicating a nontrivial, learned understanding of plasma physics.

Compared to previous studies that employed dimensionality reduction methods like t-SNE or UMAP to analyze the scenario of discharges[37–40], FusionMAE can identify whether the underlying plasma state is consistent through different combinations of diagnostic signals. As a result, the dimensionality reduction analysis no longer depends on manual feature selection or data cleaning, leading to more robust and physically meaningful representations. This is further demonstrated in Fig. 4c–e. Two discharges, shot 4046 and shot 4070, are selected from the testing set. These two shots are operated under the same template setting, resulting in highly similar plasma trajectories, as shown in Fig. 4e. However, for shot 4046, the EFIT code fails to converge during the automatic intra-shot reconstruction process and the data are therefore missing from the dataset. The D-α signal is also unavailable because the diagnostic system was not activated. Under

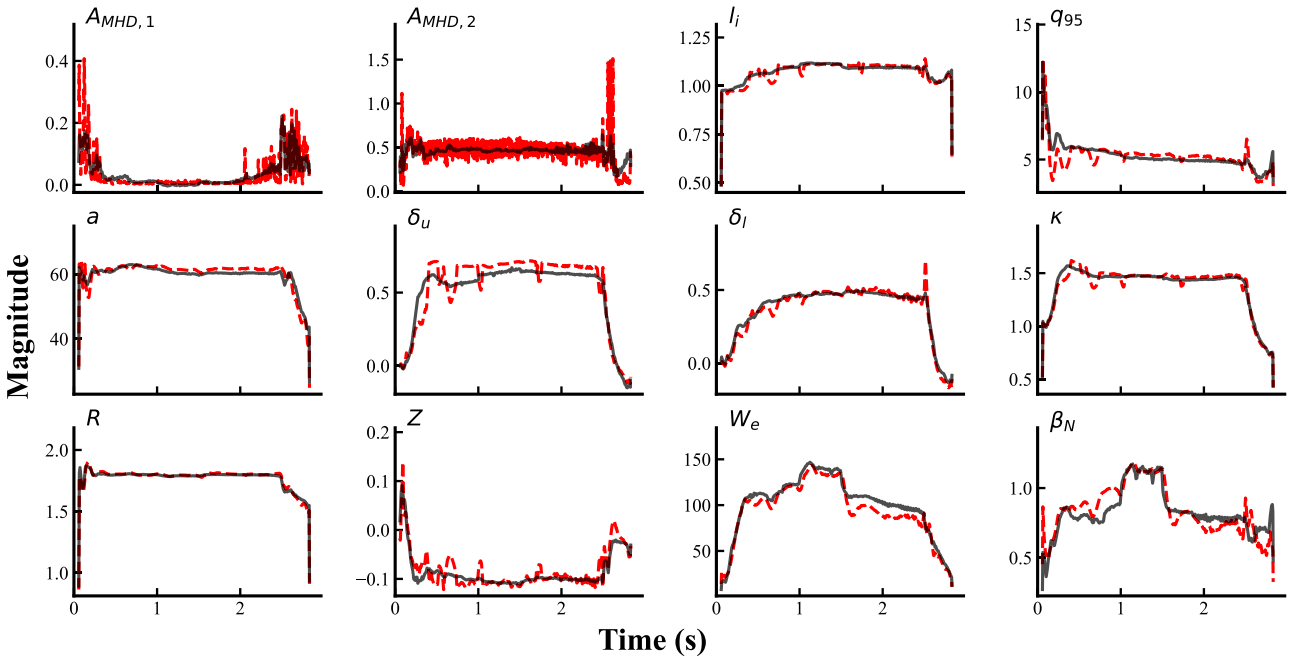



**Fig. 5 | Demonstration of FusionMAE for the automatic analysis of secondary data.** Black lines denote the reconstructed secondary signals when all of them are masked in the input. Red dashed lines represent the ground truth given by conventional analysis tools, like EFIT[41].

such circumstances, when UMAP is directly applied to the raw diagnostic signals, the trajectories of the two shots are separated due to the discrepancy in the available signal sets, as illustrated in Fig. 4d. In contrast, the UMAP projection based on the embeddings learned by FusionMAE (Fig. 4c) correctly captures the similarity between the two discharges, despite the differences in diagnostic availability.

### Downstream application: automatic data analysis for secondary data

While prior analysis has primarily emphasized FusionMAE's pretraining performance, its true value as a large-scale pretrained model lies in its broad application on downstream tasks. Some of the applications are already empirically observed and systematically characterized in this and the following two subsections.

Firstly, the capability of signal reconstruction can be reframed as a form of automated data analysis. Among the 88 input channels, some are secondary data generated by data analysis programs. For example, the plasma shape parameters ($R$, $Z$, $\kappa$, …) are calculated by the EFIT code based on the poloidal magnetic field and coil currents[41]. An extreme test case was constructed where all secondary data were masked and the result is shown in Fig.5. FusionMAE successfully reconstructed these masked secondary parameters with high fidelity, demonstrating its capacity to infer underlying physical relationships and approximate the functionality of traditional analysis tools. The statistical evaluation of FusionMAE's performance on this application is also presented in Fig. 3a, b by the blue radar plots. More detailed statistical result can be found in the column of 'Reconstructed by FusionMAE' in Supplementary Table 2.

### Downstream application: all-purpose vector for multiple control and simulation tasks

Since the plasma state embedding has integrated all the information from the 88 input signals, it is natural to be considered as an 'all-purpose vector' to support various tasks related to tokamak operation. To demonstrate this capability, three downstream tasks are selected. The first task, disruption prediction (DPR), involves identifying disruption precursors hidden in plasma parameters and exemplifies pattern recognition and anomaly detection[42–45]. The second task, EFIT-NN, is to calculate the plasma shape based on the poloidal magnetic field and coil currents, representing a data analysis challenge[46]. Instead of relying solely on FusionMAE's inherent data analysis capabilities as in Fig. 5, the shape parameters in the 88*10 input array are masked, and an individual neural network is trained to compute them based on the plasma state embedding. The third task, plasma evolution prediction (PPR), is to forecast the future plasma parameters based on the previous parameters and planned coil currents, illustrating predictive simulation[47,48]. All tasks adhere to the same problem definitions and baseline solutions as described in refs. 27–29, except for certain modifications that conflict with FusionMAE. Performance on DPR is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC)[49], while EFIT-NN and PPR are measured by the PCC between the predicted outputs and the ground truth[50]. Both AUC and PCC are bounded by 1, with values closer to 1 indicating better performance. More details about the implementation of this and the subsequent two downstream tasks can be found in Supplementary Notes 3–5

The results of comparison experiments are shown in Fig. 6 by the first 3 bars in each subfigure. In these experiments, the performances are tested for three algorithm versions using different inputs, i.e., raw data, plasma state embedding, and data with missing signals completed by FusionMAE. Impressively, both the versions using embedding and completed data achieve similar or even higher performance compared to the version using raw data. These results reveal that FusionMAE provides a precise, self-consistent representation of the plasma and benefits the downstream tasks. The superior performance of the third version over the second one likely stems from the information loss during the generation of the plasma state embedding. By combining the raw data with the reconstructed signals, the third version compensates for this loss and achieves the highest performance.

### Downstream application: robust control against missing input signals

FusionMAE provides another advantage by enabling robust tokamak operation even when some diagnostic systems are unavailable[51]. On one hand, missing signals can be replaced by mask tokens and subsequently reconstructed by FusionMAE. On the other hand, the embedded vector, which integrates information from multiple diagnostic and control systems, remains stable when some of the input signals are unavailable.

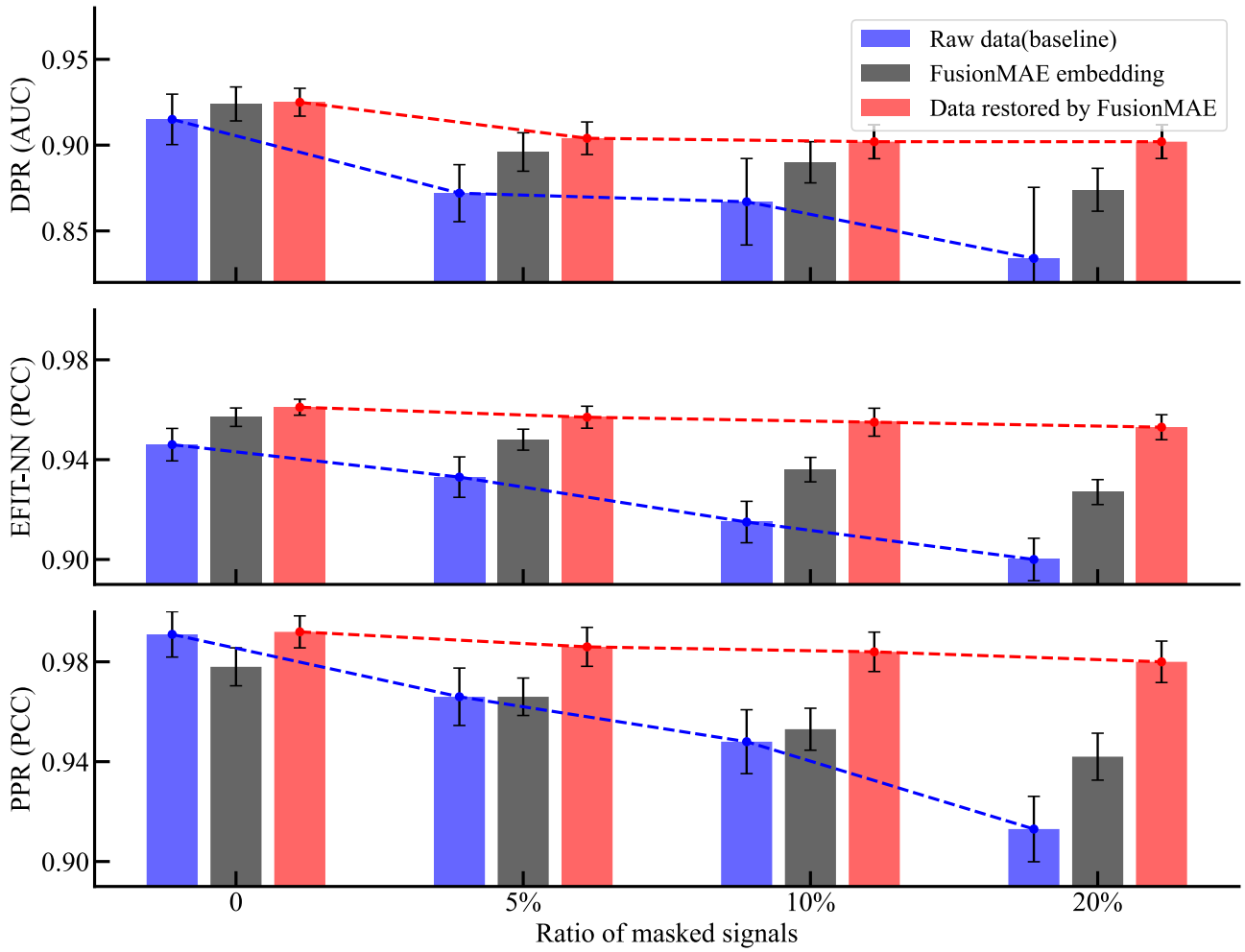



**Fig. 6 | Capability of FusionMAE to comprehensively enhance the performance of downstream algorithms.** The performance of DPR, EFIT-NN and PPR based on models utilizing the plasma state embedding and data completed by FusionMAE, compared with ordinary models that are developed on the raw data. Error bars are obtained by repeating the training and testing for 5 times and calculating the standard deviation of performance.

Figures 7a–c present the outputs of the DPR, EFIT-NN, and PPR tasks, respectively, during an HL-3 discharge, with the shaded area representing the uncertainty when 5% and 20% of the input is masked. All outputs remain relatively stable, exhibiting only minor oscillations when some input signals are masked. Figure 6 further reports the statistical performance of the DPR, EFIT-NN, and PPR tasks with 5%, 10%, and 20% of the input masked. As expected, the performance of the baseline algorithm degrades rapidly as more signals are masked, whereas the degradation for algorithms supported by FusionMAE is considerably slower. Consequently, FusionMAE establishes a robust operational framework for tokamak reactors while demonstrating potential to minimize diagnostic system requirements, which is a critical advancement toward FPP.

## Discussion

This study presents FusionMAE, a large-scale pretrained model based on the masked autoencoder (MAE) architecture[52] that captures the intrinsic correlations among diverse plasma and operational parameters in tokamaks, thereby integrating multi-source data into a unified plasma state embedding. This approach was not only validated on HL-3 but also deployed and tested on a spherical tokamak, SUNIST-2, with the results presented in the Supplementary Note 6. For fusion reactor operation, this integrated embedding shows significant potential for enhancing both system integration and reliability.

Regarding system integration, the plasma state vector generated by FusionMAE integrates information from multiple diagnostic systems, serving as a universal representation. This vector not only offers robust support for a variety of downstream computational tasks but also exhibits superior feature extraction capabilities that systematically enhance the performance of all downstream modules. Such a unified representation could revolutionize tokamak operational architecture by unifying the interface between diagnostic and control and benefitting all downstream control tasks.

In terms of operational reliability, FusionMAE tackles a critical challenge for future fusion reactors: mitigating operational instability resulting from partial diagnostic failures[53]. The model demonstrates computational resilience when handling incomplete input data, with downstream task outputs showing only bounded fluctuations and quantifiable uncertainty. Furthermore, FusionMAE achieves high-fidelity data imputation by leveraging its learned understanding of inter-parameter physical relationships, which could be used to relax the redundancy requirements for diagnostic systems for future fusion reactors.

From an AI perspective, FusionMAE represents a successful application of self-supervised pre-training paradigm[54] to fusion data science. Its emergent properties, including sophisticated feature extraction, cross-task generalization, and intelligent behaviours in data compression and imputation, confirm the applicability of pre-training paradigms like BERT[55] and GPT[56] to fusion data domains. This finding suggests that fusion-AI integration should transcend conventional small-model approaches (e.g., supervised learning) and embrace cutting-edge techniques from the large-scale model era, exploring the potential of adapting

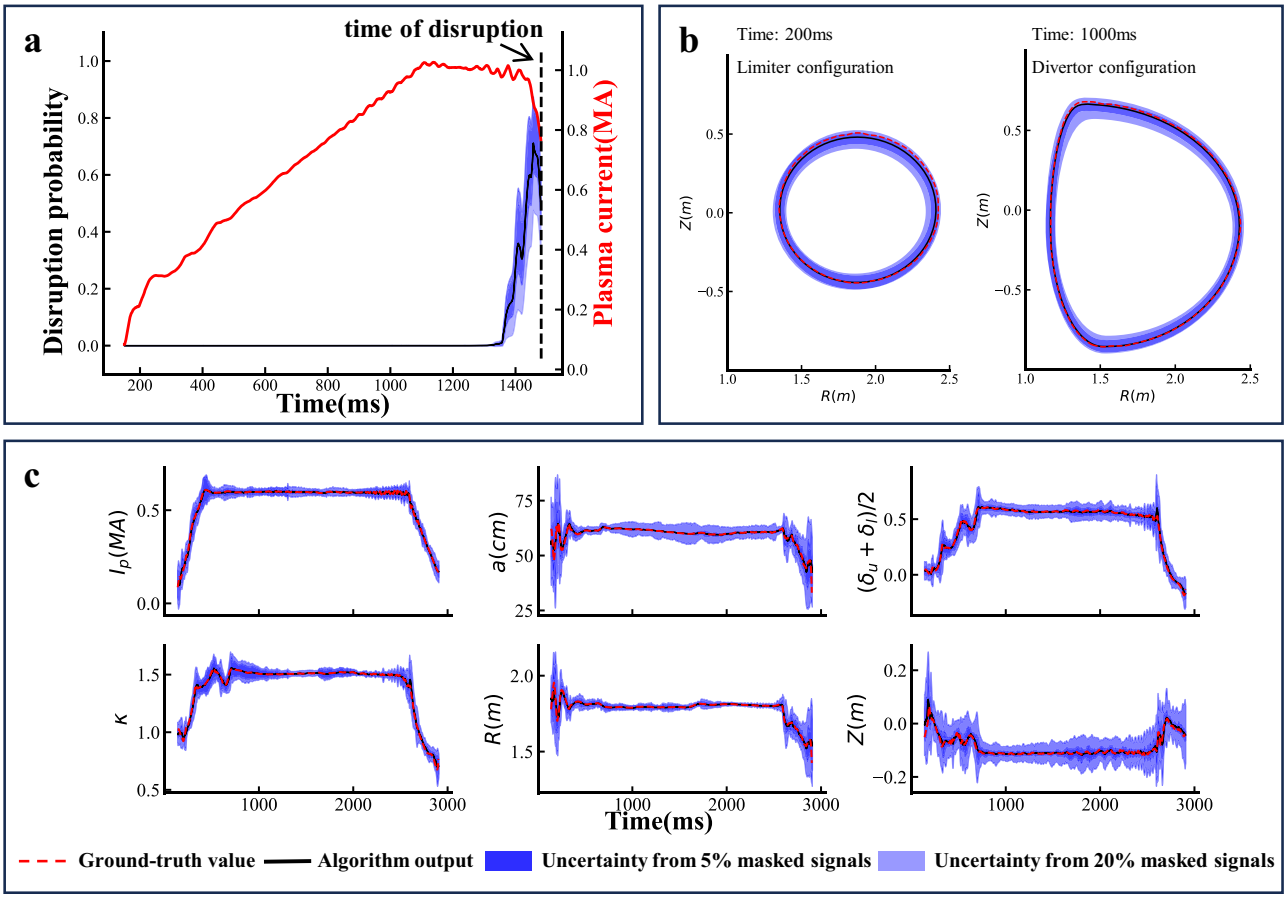



**Fig. 7 | Stable operation of downstream modules under incomplete input data, enabled by FusionMAE.** Output of DPR (**a**), EFIT-NN (**b**), PPR (**c**) model during an HL-3 discharge using the plasma state embedding. Blue bands indicate uncertainties introduced by randomly masking 5% and 20% of input signals, demonstrating robustness to missing and invalid diagnostics. The uncertainties are multiplied by 5 in (**b**, **c**) to get more visible.

state-of-the-art AI methodologies to accelerate progress toward practical fusion energy solutions.

## Methods

### Dataset description

The HL-3 Tokamak, formerly known as HL-2M, is an experimental fusion device developed by the Southwestern Institute of Physics in China. It is designed to enhance the understanding of key physics and engineering principles essential for the advancement of international fusion initiatives such as ITER. HL-3 is capable of sustaining a plasma current between 1.5 and 3 MA while operating under a magnetic field strength of 2.3 T. Its structural characteristics include a major radius of 1.78 m, a minor radius of 0.65 m, and an aspect ratio of approximately 2.8, reflecting its toroidal configuration[23,24].

The dataset utilized in this study encompasses 2,445 plasma discharges collected from four HL-3 experimental campaigns, spanning the years 2022 to 2025. The discharges are randomly divided into a development set and a test set, containing 1,950 and 495 shots, respectively. The development set is employed for training and validating the neural network, while the test set is used to assess algorithm performance. This dataset splitting strategy is consistently applied across both the pretraining and evaluation of FusionMAE, as well as the training and testing of all downstream task models. As illustrated in Fig. 2a, each discharge includes 88 recorded signals that characterize the plasma state, which can be measured in real-time in HL-3 plasma control system. To ensure alignment and streamline the FusionMAE framework, all signals are resampled at 1 kHz. The physical significance, statistical properties, and measurement methods of these signals are detailed in Table 1.

For most channels, the health rate, defined as the proportion of shots in which a given signal channel has recorded valid data, is below 100%. This observation underscores the inherent challenges and complexity of fusion diagnostics, emphasizing that missing diagnostic signals are an unavoidable reality in fusion facilities. To systematically evaluate the performance of various algorithms under different levels of missing data, we constructed three extended datasets based on the original dataset. Specifically, 5%, 10%, and 20% of the signal channels were randomly replaced with Gaussian noise curves, simulating artificial data loss. These extended datasets were subsequently used for comparative experiments, as presented in Figs. 4a, 6 and 7. The results demonstrate that FusionMAE effectively learns from incomplete datasets, overcoming a key limitation of conventional algorithms, which typically require fully clean and complete data for training and inference.

Finally, Z-score normalization is adopted for data preprocessing in FusionMAE, where each input channel is normalized using the mean and standard deviation estimated from the training set.

### Model architecture of FusionMAE

FusionMAE is a Transformer-based masked autoencoder, with its architecture and hyperparameters illustrated in Fig.2b and Table 2. The model design is inspired by the masked autoencoder for vision Transformers proposed in ref. 52. It processes an input consisting of an $88 \times 10$ array, where each row, a 10 ms time window of signals sampled at 1 kHz, corresponds to one of the 88 diagnostic channels. Within the Transformer framework, each row is treated as a patch, encapsulating plasma information from a single diagnostic channel.

To handle missing and invalid channels, a mask token is introduced in the neural network, replacing unavailable data. This

**Table 1 | The detailed information about the 88 channels used in the development of FusionMAE, including the physical significance, statistical properties and data health rate**

| Channel name | Channel number | Unit | Physical meaning | Statistical properties | | |
|---|---|---|---|---|---|---|
| | | | | Mean value | Standard deviation | Health rate |
| $I_p$ | 1 | MA | Toroidal plasma current, measured by Rogowski coils. | 0.384 | 0.228 | 100% |
| $R$ | 1 | m | Horizontal position of the plasma center in the poloidal cross-section, calculated using the EFIT code. | 1.76 | 0.110 | 90.9% |
| $Z$ | 1 | m | Vertical position of the plasma center in the poloidal cross-section, calculated using the EFIT code. | −0.0608 | 0.113 | 90.9% |
| $a$ | 1 | cm | Minor radius of the plasma in the poloidal cross-section, calculated using the EFIT code. | 58.7 | 7.68 | 90.9% |
| $\kappa$ | 1 | — | Elongation ratio of the plasma in the vertical direction, calculated using the EFIT code. | 1.22 | 0.268 | 90.9% |
| $\delta_u$ | 1 | — | Upper triangularity, defined as the degree to which the upper half of the plasma deviates from a perfectly circular shape towards a triangular form, calculated using the EFIT code. | 0.195 | 0.225 | 90.9% |
| $\delta_l$ | 1 | — | Lower triangularity, defined as the degree to which the lower half of the plasma deviates from a perfectly circular shape towards a triangular form, calculated using the EFIT code. | 0.361 | 0.405 | 90.9% |
| $B_t$ | 1 | T | Toroidal magnetic field strength at the geometric center of the vacuum vessel. | 1.31 | 0.296 | 99.7% |
| $B_p$ | 41 | a.u. | Local poloidal magnetic field strength, measured by 41 pick-up coils surrounding the plasma. | 0.0701 | 0.0777 | 99.8% |
| $V_{loop}$ | 1 | V | Toroidal loop voltage of the plasma. | 2.35 | 3.14 | 100% |
| $n_e$ | 2 | $10^{19}/m^3$ | Line-averaged electron density, measured using Faraday interferometers. | 0.943 | 0.985 | 64.4% |
| $T_e$ | 2 | keV | Electron temperature, measured using electron cyclotron emission (ECE) diagnostics. | 0.428 | 0.454 | 57.3% |
| $W_e$ | 1 | kJ | Plasma stored energy. | 67.7 | 67.8 | 90.9% |
| $P_{rad}$ | 1 | MW | Total thermal radiation power emitted by the plasma. | 0.122 | 0.194 | 63.2% |
| $SX$ | 4 | a.u. | Line-integrated soft X-ray intensity from the plasma. | 0.304 | 0.360 | 78.7% |
| $D_\alpha$ | 2 | a.u. | Intensity of the D-α spectrum. | 0.225 | 0.525 | 74.3% |
| $A_{MHD}$ | 2 | a.u. | Amplitude of magnetohydrodynamic (MHD) instabilities with a toroidal mode number $n = 1$, derived from raw signals recorded by Mirnov probes. | 0.173 | 0.292 | 93.1% |
| $A_{imp}$ | 1 | a.u. | Intensity of the C III impurity spectrum. | 0.141 | 0.270 | 64.7% |
| $\beta_N$ | 1 | — | Normalized beta, representing the normalized plasma pressure. | 0.655 | 0.525 | 90.9% |
| $q_{95}$ | 1 | — | Safety factor at the 95% flux surface. | 5.31 | 2.74 | 90.9% |
| $l_i$ | 1 | — | Internal inductance of the plasma. | 1.03 | 0.068 | 90.9% |
| $I_{PF}$ | 16 | kA | Current in the poloidal field coils of the tokamak. | -0.618 | 1.70 | 99.9% |
| $I_{CS}$ | 1 | kA | Current in the central solenoid coil of the tokamak. | 6.35 | 36.7 | 99.9% |
| $P_{NBI}$ | 1 | kW | Power of neutral beam injection. | 55.1 | 220 | 99.9% |
| $P_{EC}$ | 1 | kW | Power of electron cyclotron resonant heating. | 19.2 | 109 | 98.7% |
| $P_{LH}$ | 1 | kW | Power of lower hybrid resonant heating. | 5.10 | 30.0 | 98.6% |

mask token is a randomly initialized trainable vector, which is updated throughout the training process and gradually converges into an optimal representation that encodes missing information. In addition to these inherently missing or invalid channels, 25% of the valid channels are intentionally masked during training. These masked channels serve as the primary targets for reconstruction, enabling FusionMAE to learn to reconstruct missing information effectively, as detailed in the next section.

Afterwards, each row of input is projected into a 128-dimensional space using a multilayer perceptron (MLP), followed by the addition of a positional encoding (PE) vector[33] to encode the channel index and inform the encoder about the diagnostic source of each patch. The positional encoding is computed using Eqs. (1) and (2), where idx represents the index of the input channel, and 2i and $2i+1$ refer to the indices of the even and odd elements of the positional encoding vector, respectively.

$$\mathrm{PE}(\mathrm{idx}, 2\mathrm{i}) = \sin(\mathrm{idx}/88^{2\mathrm{i}/128})\mathrm{i} = 0, 1, 2, \ldots, 63 \quad (1)$$

$$\mathrm{PE}(\mathrm{idx}, 2\mathrm{i}+1) = \cos(\mathrm{idx}/88^{(2\mathrm{i}+1)/128})\mathrm{i} = 0, 1, 2, \ldots, 63 \quad (2)$$

The resulting $88 \times 128$ array is then passed through an encoder composed of multiple Transformer blocks, each incorporating multi-head self-attention layers[34], MLPs, layer normalizations[35], and residual connections[36]. This Transformer block architecture is widely adopted in large-scale models such as BERT and GPT, demonstrating its effectiveness in learning intricate dependencies within sequential data.

The encoder primarily captures underlying patterns in plasma behaviour. Its output maintains the same shape as the input ($88 \times 128$) and is subsequently flattened into a 11264-dimensional vector. This vector is

**Table 2 | The detailed information about model structure of FusionMAE**

| Number | Repeat Time | Layer type | Output shape | Layer parameters |
|---|---|---|---|---|
| 1 | 1 | Input layer | 88*10 | |
| 2 | 1 | Multi-layer perceptron | 88*128 | 128 units, dropout rate = 0.2 |
| 3 | 20 | LayerNorm | 88*128 | |
| 4 | | Multi-head attention | 88*128 | 8 heads, key dimension = 128, dropout rate = 0.2 |
| 5 | | LayerNorm | 88*128 | |
| 6 | | Multi-layer perceptron | 88*256 | 256 units, dropout rate = 0.2 |
| 7 | | Multi-layer perceptron | 88*128 | 128 units, dropout rate = 0.2 |
| 8 | 1 | Reshape | 11,264 | |
| 9 | 1 | Multi-layer perceptron | 256 | 256 units, dropout rate = 0.2 |
| 10 | 1 | Multi-layer perceptron | 11264 | 11264 units, dropout rate = 0.2 |
| 11 | 1 | Reshape | 88*128 | |
| 12 | 4 | LayerNorm | 88*128 | |
| 13 | | Multi-head attention | 88*128 | 8 heads, key dimension = 128, dropout rate = 0.2 |
| 14 | | LayerNorm | 88*128 | |
| 15 | | Multi-layer perceptron | 88*256 | 256 units, dropout rate = 0.2 |
| 16 | | Multi-layer perceptron | 88*128 | 128 units, dropout rate = 0.2 |
| 17 | 1 | Multi-layer perceptron | 88*10 | 10 units, dropout rate = 0.2 |

further compressed by an MLP into a 256-dimensional embedded representation.

The decoder adopts a structure that closely mirrors the encoder. The 256-dimensional embedded vector is first expanded back into a 11264-dimensional vector, which is then reshaped into an $88 \times 128$ array. Subsequently, multiple Transformer blocks process this array to reconstruct the original data.

The key distinction between the encoder and decoder lies in the treatment of positional encoding. Unlike in the encoder, where positional embeddings are directly added to the input patches, in decoder, they serve as query vectors in the final multi-head attention layer, aiding in the reconstruction process.

Finally, an MLP layer projects the $88 \times 128$ array into an $88 \times 10$ array, representing the reconstructed diagnostic signals across all channels, including those that were originally missing or invalid.

### Training of FusionMAE

The FusionMAE model is trained using the AdamW optimizer, aiming to minimize the loss function defined in Eq. (3). In this formulation, $X_{recontruct}$ represents the output of FusionMAE. $X_{original}$ denotes the original input data. Term W assigns weights to each input channel, with values set as 0.2, 0, and 1 for valid channels, invalid or missing channels, and intentionally masked channels, respectively. N represents the number of samples in a training batch.

$$\text{loss} = \frac{1}{N}\sum_{1}^{N}\sum_{1}^{88} W * (X_{recontruct} - X_{original})^2 \quad (3)$$

This weighting strategy ensures that FusionMAE primarily learns to reconstruct the intentionally masked channels, while simultaneously preserving valid information during the compression process. Given the model's architecture, its capability to reconstruct intentionally masked channels naturally extends to the reconstruction of invalid and missing channels, as these are treated equivalently during training.

A systematic set of ablation experiments is conducted to determine the optimal hyperparameters of FusionMAE, with the results summarized in Fig. 8.

Model depth is firstly investigated by varying the total number of Transformer blocks. As shown in Fig. 8a, FusionMAE exhibits progressively improved and more stable performance as the network depth increases, with the performance gain saturating beyond 24 stacked blocks. Based on this observation, the total depth is fixed at 24 layers. We further examine the placement of the bottleneck layer in Fig. 8b and observe a modest but consistent improvement when the bottleneck is shifted toward the output end, corresponding to a deeper encoder and a shallower decoder. Additional scans, including varying the encoder depth with a fixed 4-layer decoder (Fig. 8c) and varying the decoder depth with a fixed 20-layer encoder (Fig. 8d), yield consistent trends. These results suggest that the encoder bears a higher modeling burden, as it must accommodate diverse patterns of missing data, whereas the decoder mainly performs a relatively fixed reconstruction process based on the learned plasma embedding, which has largely disentangled the influence of missing signals.

After fixing the model depth, we explore the model width. The first width-related parameter is the dimensionality of the bottleneck layer (Fig. 8e), which also defines the dimension of the plasma state embedding. This dimension must be smaller than the total input dimensionality (880) to prevent tricky solutions such as direct memorization of the inputs, while still retaining sufficient expressive power to represent the wide range of plasma states. Scanning this parameter from 64 to 1024 reveals that a dimension of 256 achieves the best performance on the HL-3 dataset. Another width parameter is the patch projection dimension (Fig.8f), which controls the internal width of the Transformer blocks. Increasing this dimension leads to improved and more stable performance; however, it also results in a linear increase in model size. A value of 128 is therefore selected as a balanced choice, providing strong performance with an acceptable computational cost.

In its current form, FusionMAE does not incorporate a dedicated architectural treatment for temporal dynamics. As a result, model performance degrades noticeably when the patch length becomes too long. The experiments in Fig. 8g indicate that this transition occurs at approximately 10 ms. Addressing this limitation would likely require, on the one hand, augmenting the base Transformer with mechanisms tailored for time-series modeling, and on the other hand, potentially reformulating the pretraining framework in an autoregressive manner.

For the ratio of intentionally masked patches in Fig. 8h, the results provide particularly informative insights. When the masking ratio remains below a critical threshold (approximately 30%), the testing loss remains nearly constant, indicating that FusionMAE effectively exploits cross-channel correlations and reflecting a substantial level of redundancy in the

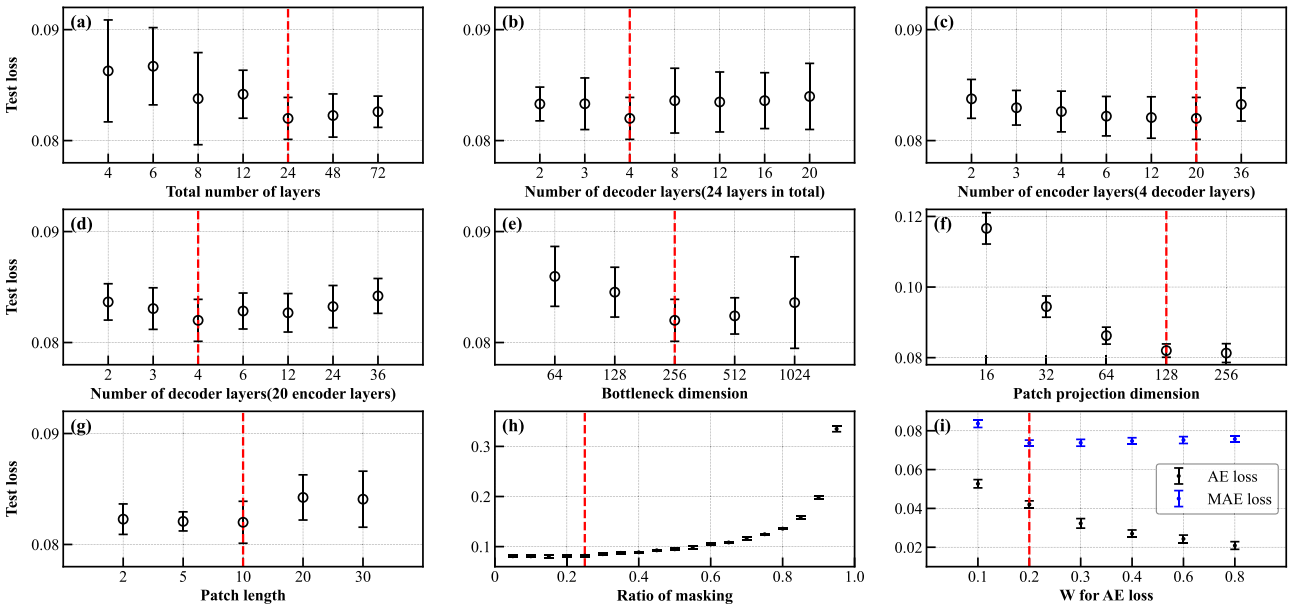



**Fig. 8 | Results of comparison experiments used to specify the hyperparameter settings of FusionMAE.** The tuned parameters are (**a**) number of total layers, (**b**) number of decoder layers with fixed total layers, (**c**) number of encoder layers with fixed decoder layers, (**d**) number of decoder layers with fixed encoder layers, (**e**) dimension of bottleneck layer, (**f**) projection dimension of each patch of data, (**g**) input patch length, (**h**) intentionally masked ratio, (**i**) weight for AE loss term.

diagnostic measurements. Beyond this threshold, the loss increases in a nonlinear manner, revealing a fundamental information limit where plasma state reconstruction becomes underdetermined. This behavior establishes a quantitative framework for assessing diagnostic redundancy and provides empirical guidance for identifying minimum viable diagnostic configurations, which is especially relevant for balancing measurement robustness and system complexity in future fusion reactors.

The final hyperparameter to determine is $W$ in Eq. (3). As shown in Fig. 8i, increasing $W$ steadily reduces the loss for valid data (AE loss). However, beyond $W = 0.2$, the loss for masked data (MAE loss) begins to increase. Since our study prioritizes FusionMAE's robustness to missing diagnostics and its ability to capture inter-diagnostic correlations, we selected $W = 0.2$, corresponding to the minimum MAE loss.

Other training-related hyperparameters are summarized in the Supplementary Table 4. The final FusionMAE model contains approximately 14 million trainable parameters. Training requires about 10 h on a single NVIDIA A100 GPU, while inference for a single time slice takes 4.8 ms, enabling quasi-realtime application when combined with a 10 ms input time window.

Overall, the training of FusionMAE demonstrates high repeatability and low sensitivity to hyperparameter choices. This behavior stands in marked contrast to previously developed task-specific small models for HL-3 and highlights the robustness of self-supervised pretraining combined with Transformer-based architecture. These results suggest that such a framework may serve as a key foundation for next-generation AI-for-fusion solutions.

## Data availability

The data behind the figures in this paper are provided in Supplementary Data 1. And the data used to train and test the models in this study are stored in the HL-3 experimental database. Access requires a formal application to the Southwestern Institute of Physics, which can be facilitated through avenues such as joint experimental proposals or collaborative research projects. For specific inquiries, please contact the corresponding author, Wulyu Zhong. Some data from SUNIST-2 are also included in the Supplementary Note 6. Access to these data requires contacting Tsinghua University and the corresponding author, Zhe Gao.

## Code availability

The code that supports the findings of this study is available from the first author Zongyu Yang and corresponding author, Wulyu Zhong upon request for academic research purposes. We warmly welcome collaborations, particularly those focused on the development of a cross-device FusionMAE.

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## References

1. Maggi, C. F. et al. Overview of T and D–T results in JET with ITER-like wall. *Nucl. Fusion* **64**, 112012 (2024).
2. Han, H. et al. A sustained high-temperature fusion plasma regime facilitated by fast ions. *Nature* **609**, 269–275 (2022).
3. Song, Y. et al. Realization of thousand-second improved confinement plasma with Super I-mode in tokamak EAST. *Sci. Adv.* **9**, eabq5273 (2023).
4. Bigot, B. Preparation for assembly and commissioning of ITER. *Nucl. Fusion* **62**, 042001 (2022).
5. Fasoli, A. Essay: Overcoming the Obstacles to a Magnetic Fusion Power Plant. *Phys. Rev. Lett.* **130**, 220001 (2023).
6. Shimada, M. et al. Progress in the ITER physics basis—chapter 1: overview and summary. *Nucl. Fusion* **47**, S1 (2007).
7. Costley, A. E. et al. Towards Diagnostics for a Fusion Reactor. *IEEE Trans. Plasma Sci.* **38**, 2934–2943 (2010).
8. Fasoli, A. et al. Computational challenges in magnetic-confinement fusion physics. *Nat. Phys.* **12**, 411–423 (2016).
9. Kurihara, K. et al. Plasma control systems relevant to ITER and fusion power plants. *Fusion Eng. Des.* **83**, 959–970 (2008).
10. Donné A. J. H. et al. Progress in the ITER physics basis—chapter 7: diagnostics. Nuclear Fusion 47, S337 (2007).
11. Imbeaux, F. et al. Design and first applications of the ITER integrated modelling & analysis suite. *Nucl. Fusion* **55**, 123006 (2015).
12. Gribov, Y. et al. Progress in the ITER physics basis—chapter 8: Plasma operation and control. *Nucl. Fusion* **47**, S385 (2007).
13. Bigot, B. ITER construction and manufacturing progress toward first plasma. *Fusion Eng. Des.* **146: A,**, 124–129 (2019).

14. Krizhevsky, A., Sutskever, I. & Hinton, G. E. ImageNet classification with deep convolutional neural networks. *Commun. ACM* **60**, 84–90 (2017).
15. Floridi, L. & Chiriatti, M. GPT-3: Its Nature, Scope, Limits, and Consequences. *Minds Mach.* **30**, 681–694 (2020).
16. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
17. Li, J. et al. AI Applications through the Whole Life Cycle of Material Discovery. *Matter* **3**, 393–432 (2020).
18. Bi, K. et al. Accurate medium-range global weather forecasting with 3D neural networks. *Nature* **619**, 533–538 (2023).
19. Kates-Harbeck, J., Svyatkovskiy, A. & Tang, W. Predicting disruptive instabilities in controlled fusion plasmas through deep learning. *Nature* **568**, 526–531 (2019).
20. Vega, J. et al. Disruption prediction with artificial intelligence techniques in tokamak plasmas. *Nat. Phys.* **18**, 741–750 (2022).
21. Degrave, J. et al. Magnetic control of tokamak plasmas through deep reinforcement learning. *Nature* **602**, 414–419 (2022).
22. Seo, J. et al. Avoiding fusion plasma tearing instability with deep reinforcement learning. *Nature* **626**, 746–751 (2024).
23. Duan, X. R. et al. Progress of HL-2A experiments and HL-2M program. *Nucl. Fusion* **62**, 042020 (2022).
24. Duan, X. R. et al. Recent advance progress of HL-3 experiments. *Nucl. Fusion* **64**, 112021 (2024).
25. Han, K. et al. A Survey on Vision Transformer. *IEEE Trans. Pattern Anal. Mach. Intell.* **45**, 87–110 (2023).
26. Yang, L. & Yang, T. Word embedding for understanding natural language: a survey. *Guide to big data applications*, 83–104, (2018)
27. Yang, Z. et al. Implementing deep learning-based disruption prediction in a drifting data environment of new tokamak: HL-3. *Nucl. Fusion* **65**, 026030 (2025).
28. Zheng, G. H. et al. Real-time equilibrium reconstruction by multi-task learning neural network based on HL-3 tokamak. *Nucl. Fusion* **64**, 126041 (2024).
29. Wu, N. et al. High-Fidelity Data-Driven Dynamics Model for Reinforcement Learning-based Magnetic Control in HL-3 Tokamak. *Commun. Phys.* **8**, 393 (2025).
30. Huang, M. et al. Design and Research of Electron Cyclotron Resonance Heating and Current Dive System on HL-2M Tokamak. *EPJ Web Conf.* **147**, 04006 (2017).
31. Ma, R. et al. Eddy current effect study on the HL-3 plasma equilibrium reconstruction. *Fusion Eng. Des.* **212**, 114816 (2025).
32. Wang, A. et al. Final development and preliminary experiment progress of in-vessel resonant magnetic perturbation coils system on HL-3 tokamak. *Fusion Eng. Des.* **208**, 114702 (2024).
33. Chen, P.-C. et al. A Simple and Effective Positional Encoding for Transformers. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* 2974–2988 (2021).
34. Vaswani A., et al. Attention is all you need. *Advances in Neural Information Processing Systems*, **30**, (2017).
35. Ioffe, S. & Szegedy, C. *Proceedings of the 32nd International Conference on Machine Learning*, **37**, 448–456, (2015).
36. He K. et al. *Deep residual learning for image recognition. Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778, (2016).
37. Pau, A. et al. A machine learning approach based on generative topographic mapping for disruption prevention and avoidance at JET. *Nucl. Fusion* **59**, 106017 (2019).
38. Aymerich, E. et al. A statistical approach for the automatic identification of the start of the chain of events leading to the disruptions at JET. *Nucl. Fusion* **61**, 036013 (2021).
39. Zhu, J. et al. Scenario adaptive disruption prediction study for next generation burning-plasma tokamaks. *Nucl. Fusion* **61**, 114005 (2021).
40. Wei, Y. et al. A dimensionality reduction algorithm for mapping tokamak operational regimes using a variational autoencoder (VAE) neural network. *Nucl. Fusion* **61**, 126063 (2021).
41. Sun, X. et al. Impact of various DIII-D diagnostics on the accuracy of neural network surrogates for kinetic EFIT reconstructions. *Nucl. Fusion* **64**, 086065 (2024).
42. Yang, Z. et al. A disruption predictor based on a 1.5-dimensional convolutional neural network in HL-2A. *Nucl. Fusion* **60**, 016017 (2020).
43. Tobin, M. et al. Vertical instability forecasting and controllability assessment of multi-device tokamak plasmas in DECAF with data-driven optimization. *Plasma Phys. Controlled Fusion* **66**, 105020 (2024).
44. Zheng, W. et al. Disruption prediction for future tokamaks using parameter-based transfer learning. *Commun. Phys.* **6**, 181 (2023).
45. Zhu, J. X. et al. Hybrid deep-learning architecture for general disruption prediction across multiple tokamaks. *Nucl. Fusion* **61**, 026007 (2021).
46. Qian, J. P. et al. EAST equilibrium current profile reconstruction using polarimeter-interferometer internal measurement constraints. *Nucl. Fusion* **57**, 036008 (2016).
47. Wan, C. et al. A machine-learning-based tool for last closed-flux surface reconstruction on tokamaks. *Nucl. Fusion* **63**, 056019 (2023).
48. Kim, S. H. et al. Full tokamak discharge simulation of ITER by combining DINA-CH and CRONOS. *Plasma Phys. Controlled Fusion* **51**, 105007 (2009).
49. Bewick, V., Cheek, L. & Ball, J. Statistics review 13: Receiver operating characteristic curves. *Crit. Care* **8**, 508 (2004).
50. Cohen, I. et al. Pearson correlation coefficient. *Noise reduction in speech processing*, 1–4 (2009).
51. Walsh, M. et al. Integration of diagnostics on ITER. *IEEE 26th Symposium on Fusion Engineering* 1–8 (2015).
52. He, K. et al. Masked autoencoders are scalable vision learners. *Proceedings of the* IEEE/CVF conference on computer vision and pattern recognition. 16000–16009, (2022).
53. Mitarai, O. & Muraoka, K. Analyses of Diagnostic Failure Effect and Fail-Safe Ignited Operation in a Tokamak Fusion Reactor. *Fusion Technol.* **36**, 194–211 (1999).
54. Goyal, S. et al. Finetune like you pretrain: Improved finetuning of zero-shot vision models. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 19338–19347, (2023).
55. Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Preprint at http://arXiv.org/abs/1810.04805 (2019).
56. Radford, A., Narasimhan, K., Salimans, T. & Sutskever, I. Improving language understanding by generative pre-training. Preprint at http://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf (2018).


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## Author contributions

Zongyu Yang contributed by proposing the idea, designing the algorithm, conducting coding and testing, exploring the downstream tasks, discussing the experimental results, adapting the model to SUNIST-2, writing and

revising the manuscript. Zhenghao Yang contributed by performing coding and testing and preparing key result figures. Wenjing Tian contributed by performing coding and testing, reorganizing the experimental results from the perspective of large-scale pretrained model, adapting the model to SUNIST-2, and writing the manuscript. Jiyuan Li contributed by exploring the downstream task of plasma evolution prediction and adapting the model to SUNIST-2. Xiang Sun contributed by exploring the application of FusionMAE on accessing data quality. Guohui Zheng contributed by exploring the downstream task of plasma equilibrium fitting. Songfen Liu contributed by joint supervising the downstream task of plasma equilibrium fitting. Niannian Wu contributed by exploring the downstream task of plasma evolution prediction. Rongpeng Li contributed by joint supervising the downstream task of plasma evolution prediction. Zhaohe Xu contributed by exploring the downstream task of disruption prediction. Bo Li contributed by joint supervising the development of three downstream algorithms. Zhongbing Shi contributed to diagnostic systems. Zhe Gao contributed by supervising the development of FusionMAE, writing the manuscript and discussing the experimental results. Wei Chen contributed by discussing the experimental results. Xiaoquan Ji contributed to the experiments. Min Xu contributed by writing the manuscript and discussing the experimental results. Wulyu Zhong contributed by supervising the development of FusionMAE, writing the manuscript, discussing the experimental results, and managing project progress.

## Competing interests

The authors declare no competing interests.

## Additional information

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